

STAGE 4 GEOGRAPHY

Landforms and Landscapes

REVISION DECK

Restyled in SPCC house style from the original Topic 2 deck (2017), keeping its photos, diagrams and sources.

Learning intention

We are learning how natural processes shape the Earth's landscapes, and why those landscapes matter to people.

Success criteria - I can:

- **I can explain weathering, erosion, deposition and plate tectonics.**
- **I can describe the aesthetic, cultural and economic value of a landscape.**
- **I can explain how people change landscapes, using the Australian coast as an example.**

What processes and impacts are occurring to these landscapes?

- Every landscape you will revise in this deck has been shaped by natural processes over a very long time.
- As you work through each case study, keep asking: what process made this, and what does it mean for the people who live there?



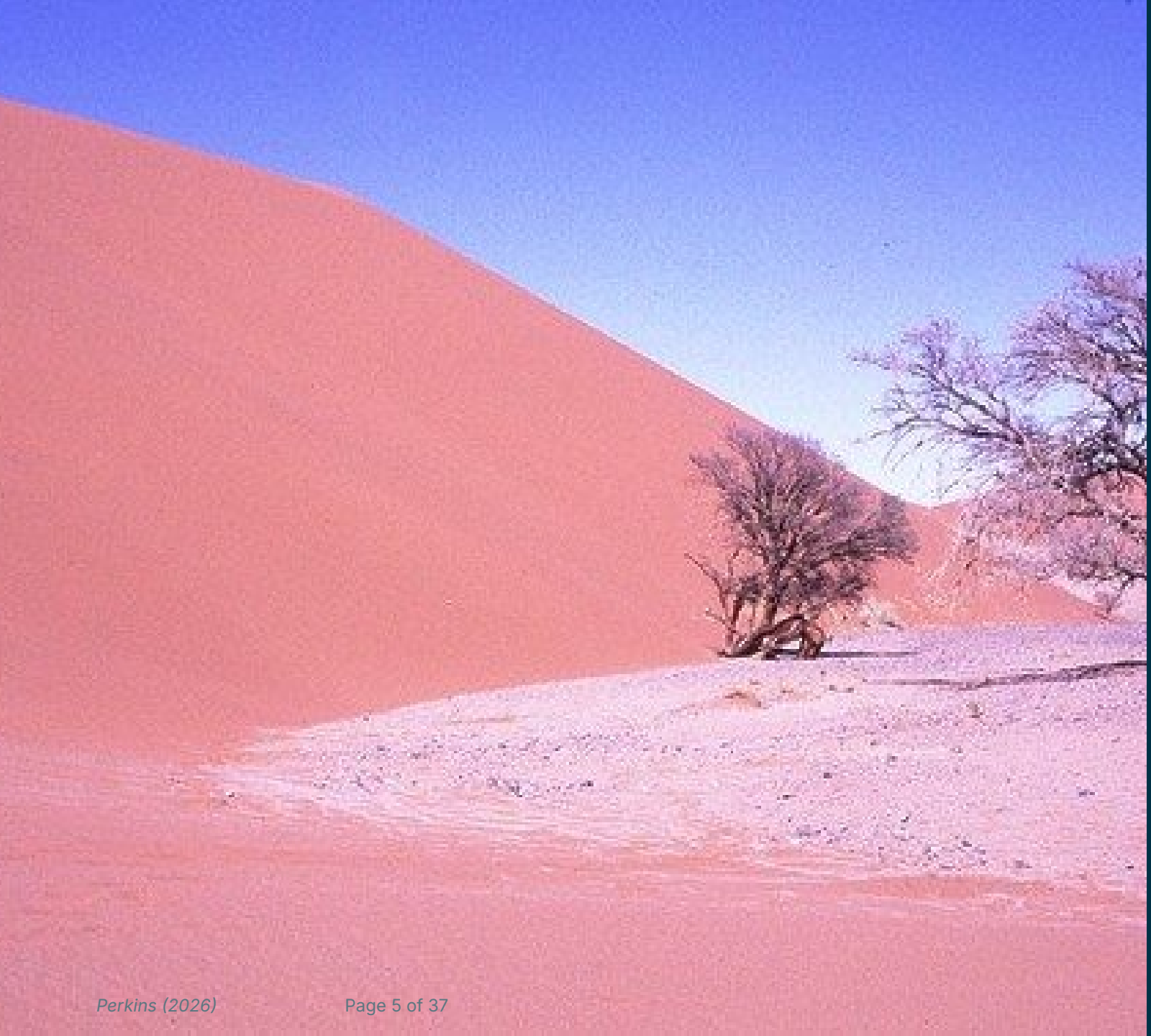
Image: NASA, "The Blue Marble" (Apollo 17, 1972) - public domain.



COASTLINE LANDSCAPE

Sydney, Australia

Image: original resource, source listed in references.



ARID LANDSCAPE

Namibia

Image: original resource, source listed in references.



MOUNTAIN LANDSCAPE

The Alps, Switzerland

Image: original resource, source listed in references.



RAINFOREST LANDSCAPE

Borneo, Malaysia

Image: original resource, source listed in references.



RIVERINE LANDSCAPE

The Nile, Egypt

Image: original resource, source listed in references.



KARST LANDSCAPE

South China

Image: original resource, source listed in references.

Geomorphic processes

Weathering, erosion, deposition and plate tectonics

The processes that create landforms

Geomorphic processes are natural processes that transform the lithosphere to create distinctive landscapes and landforms.

1

WEATHERING

The breakdown of rocks and sediments into smaller particles or a solution. Occurs in a range of landscapes, e.g. coastlines and mountains.

2

EROSION

The wearing away of the earth's surface, after weathering. Affected by the actions of wind and water.

3

DEPOSITION

The relocation of eroded matter into a new location. Rivers and coastline landscapes have a large amount of deposition.

4

PLATE TECTONICS

The movement of the major plates that make up the earth's crust, between continents and the ocean floor. Types: divergent, collision, convergent or transform.

Erosion in action: a house on the edge

- This house shows what erosion looks like when it is left unmanaged.
- Wind and water have worn away the cliff underneath it over many years, and the land it once stood well back from is now gone.
- This is also a liveability and planning issue - not just a geomorphic one.



Image: original resource, source listed in references.

Over to you

Look at the diagram below. List the processes that are occurring on this coastline.

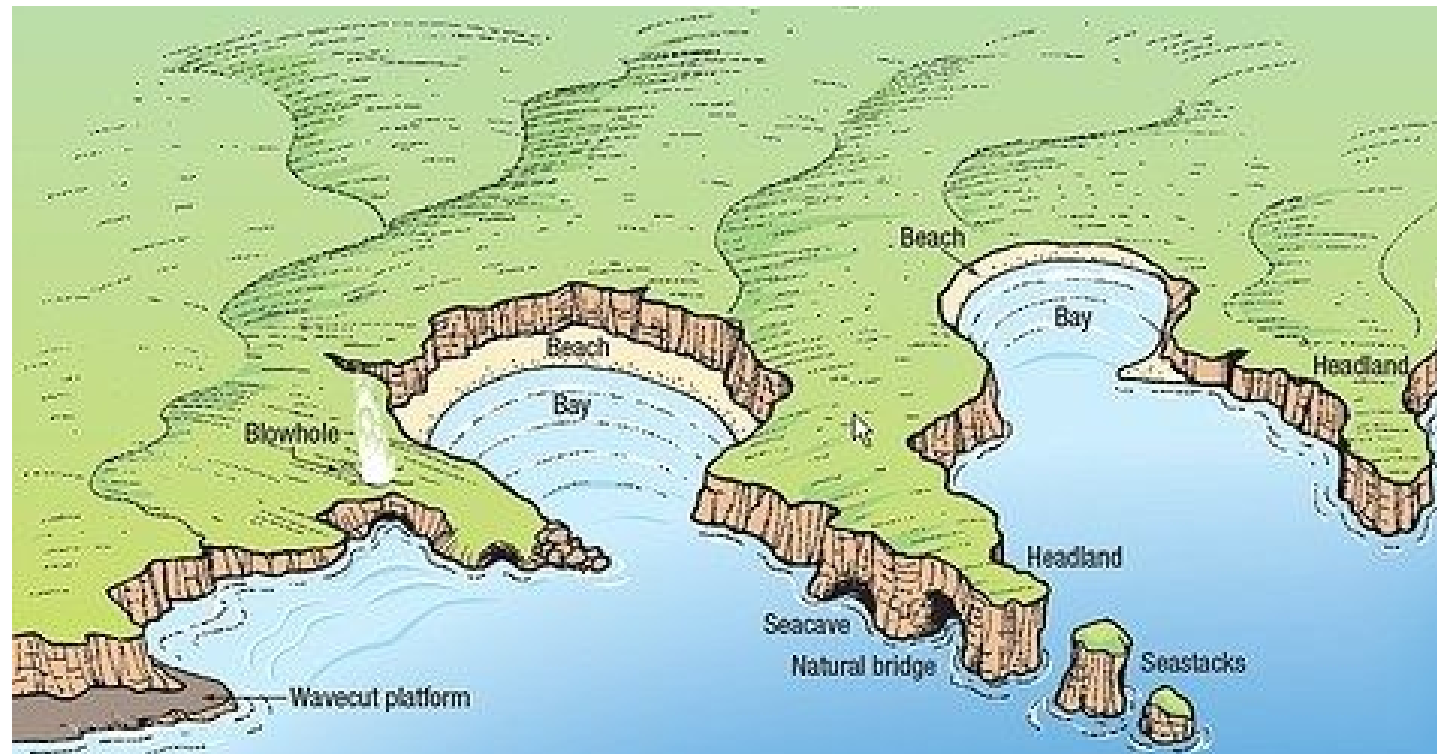


Diagram: features of coasts caused by erosion - original resource, source listed in references.

Think about headlands, bays, wave-cut platforms, blowholes and sea stacks.

Deposition: building new land

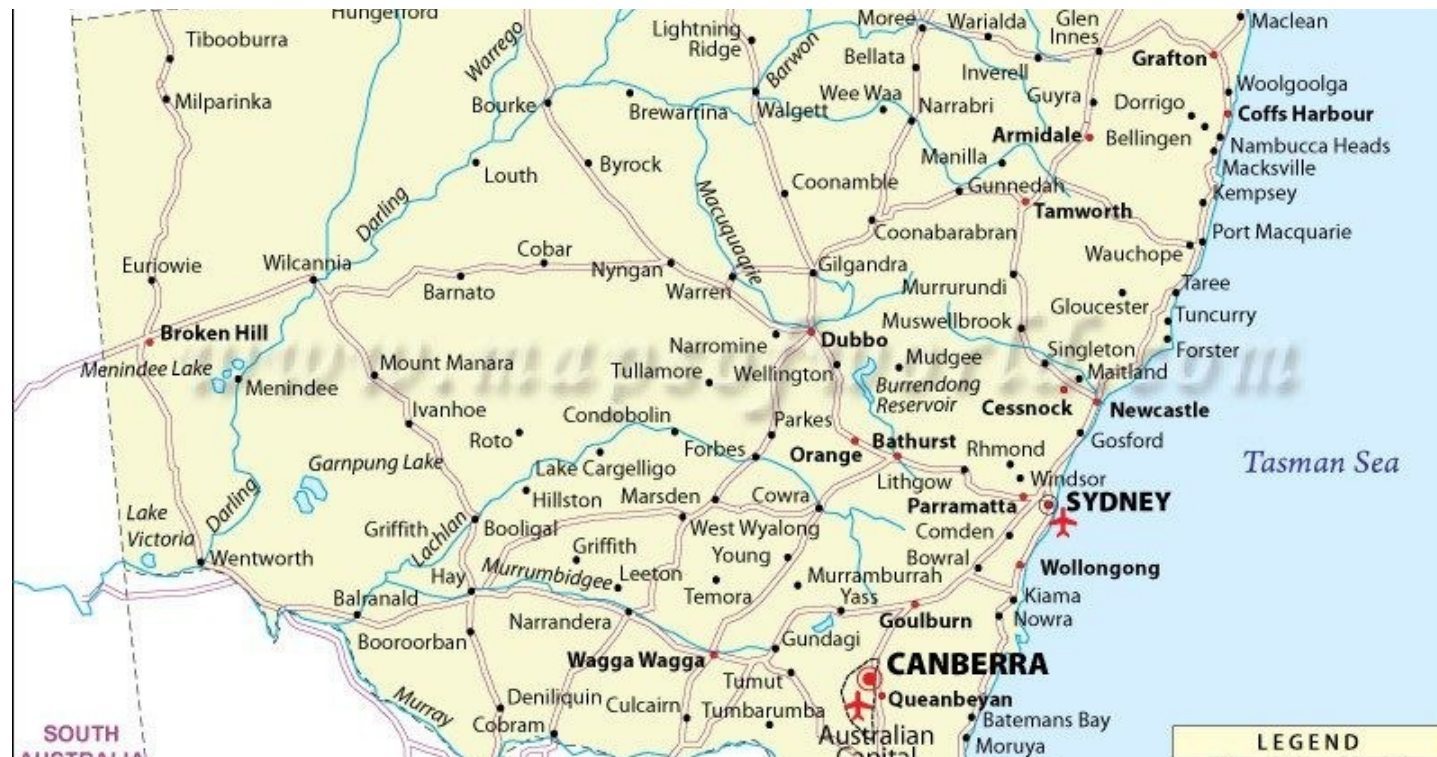
- Deposition is the relocation of eroded matter into a new location.
- Rivers and coastline landscapes have a large amount of deposition - this river delta shows sediment building up new land where the river meets the sea.



Image: original resource, source listed in references.

Over to you

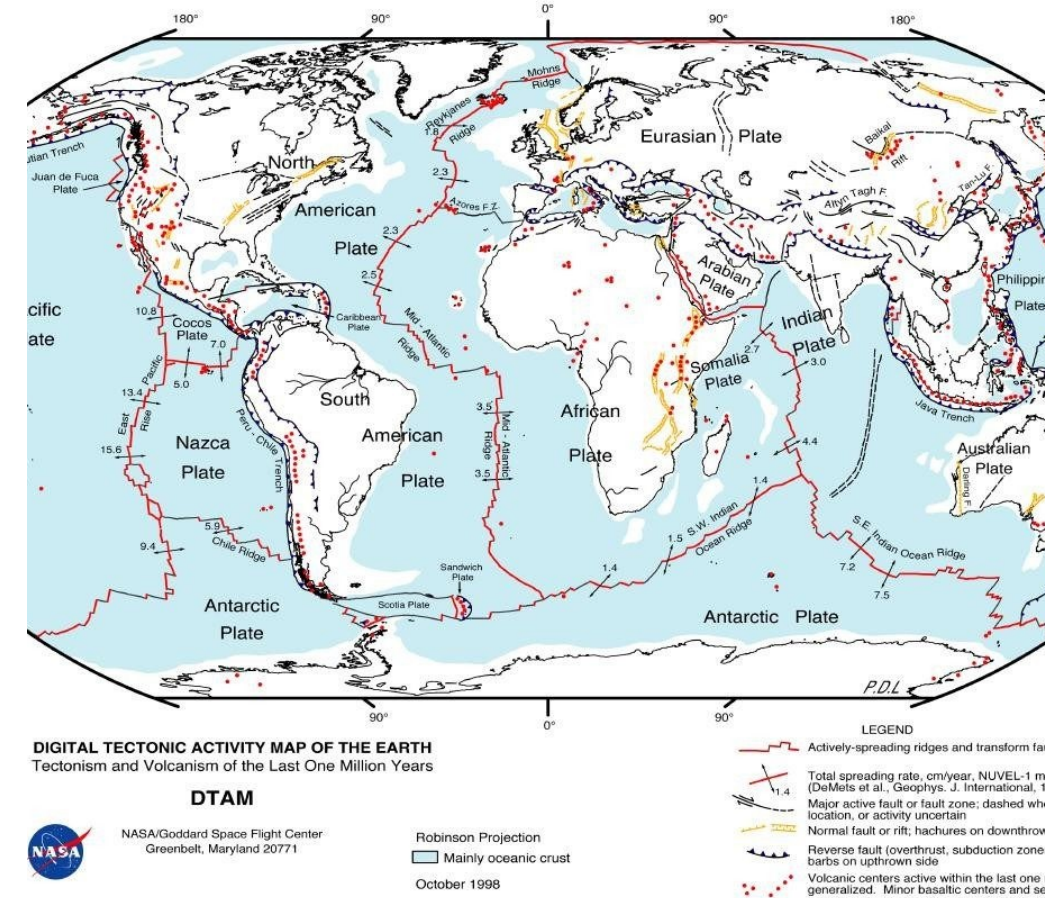
Using Google Maps on Google Drive, make a map of the major rivers in NSW that flow to the Pacific Ocean. Your map needs a title, labels of the rivers you find, and you need to measure one river.



Map: mapsofworld.com - original resource, source listed in references.

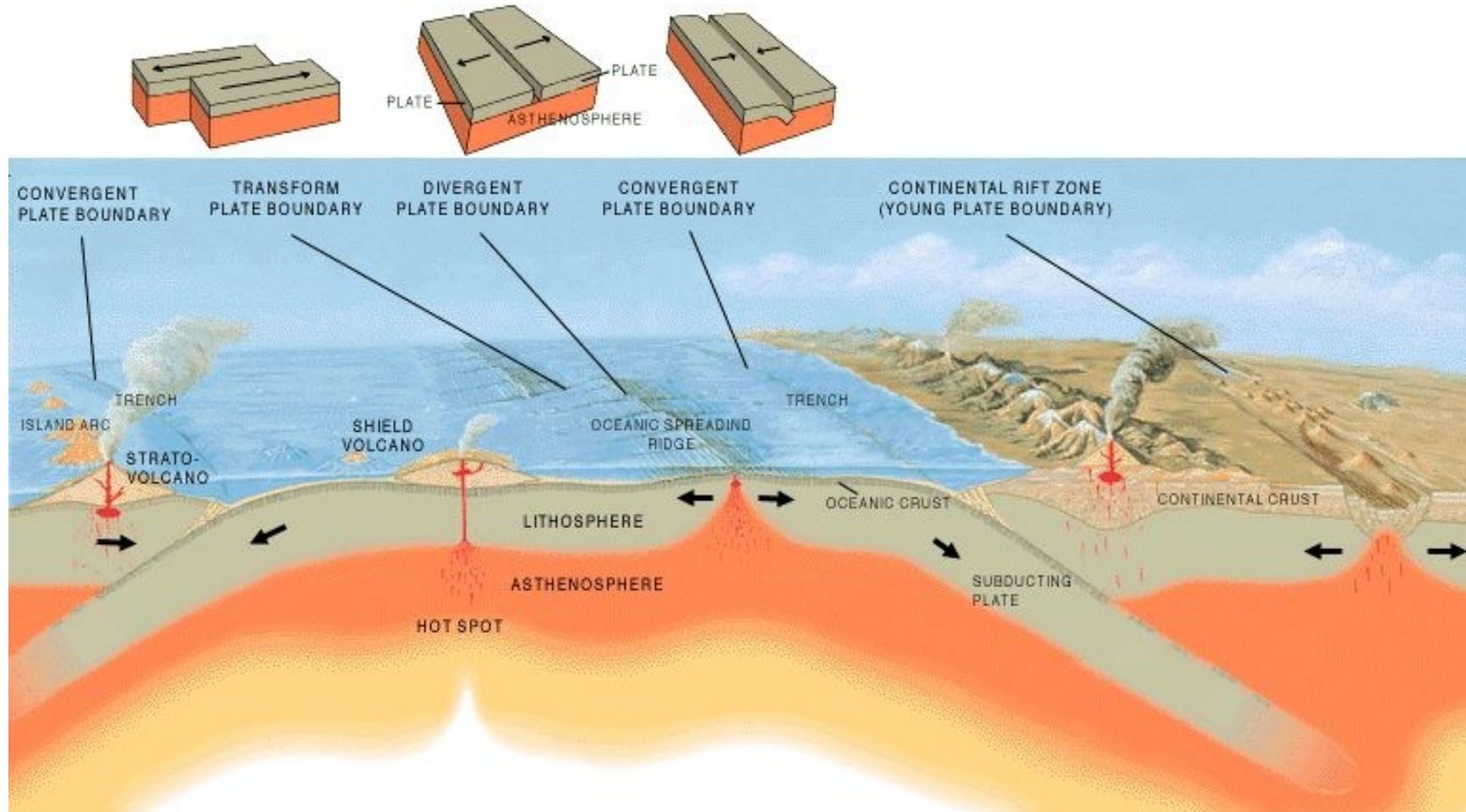
Plate tectonics

- The movement of the major plates that make up the earth's crust.
- These movements occur between continents and the ocean floor.
- Types of processes: divergent, collision, convergent or transform boundaries.



Map: NASA Goddard Space Flight Center, Digital Tectonic Activity Map of the Earth.

How plate boundaries move



Convergent, divergent and transform plate boundaries.

Diagram: original resource, source listed in references.

Watch: landscapes shaped by weathering and erosion

[Watch: Landscapes and Landforms: Coastal landscapes \(ABC Education\)](#)

Watch: an introduction to plate tectonics

[Watch: Introduction to plate tectonics \(Khan Academy\)](#)

Aesthetic and cultural value

Why landscapes matter to culture and identity

Aesthetic and cultural value

Aesthetic value is the value placed on something due to its beauty. Cultural value is the emphasis placed on something for its importance and place in society.

1

BEAUTY

Landscapes can be valued simply for how they look - their colours, scale and form.

2

IDENTITY

Some landscapes become part of a group's sense of who they are.

3

SPIRITUAL MEANING

Many landscapes hold deep spiritual or cultural significance for the people connected to them.

4

HERITAGE LISTING

The most significant sites can be World Heritage listed to protect their value for the future.

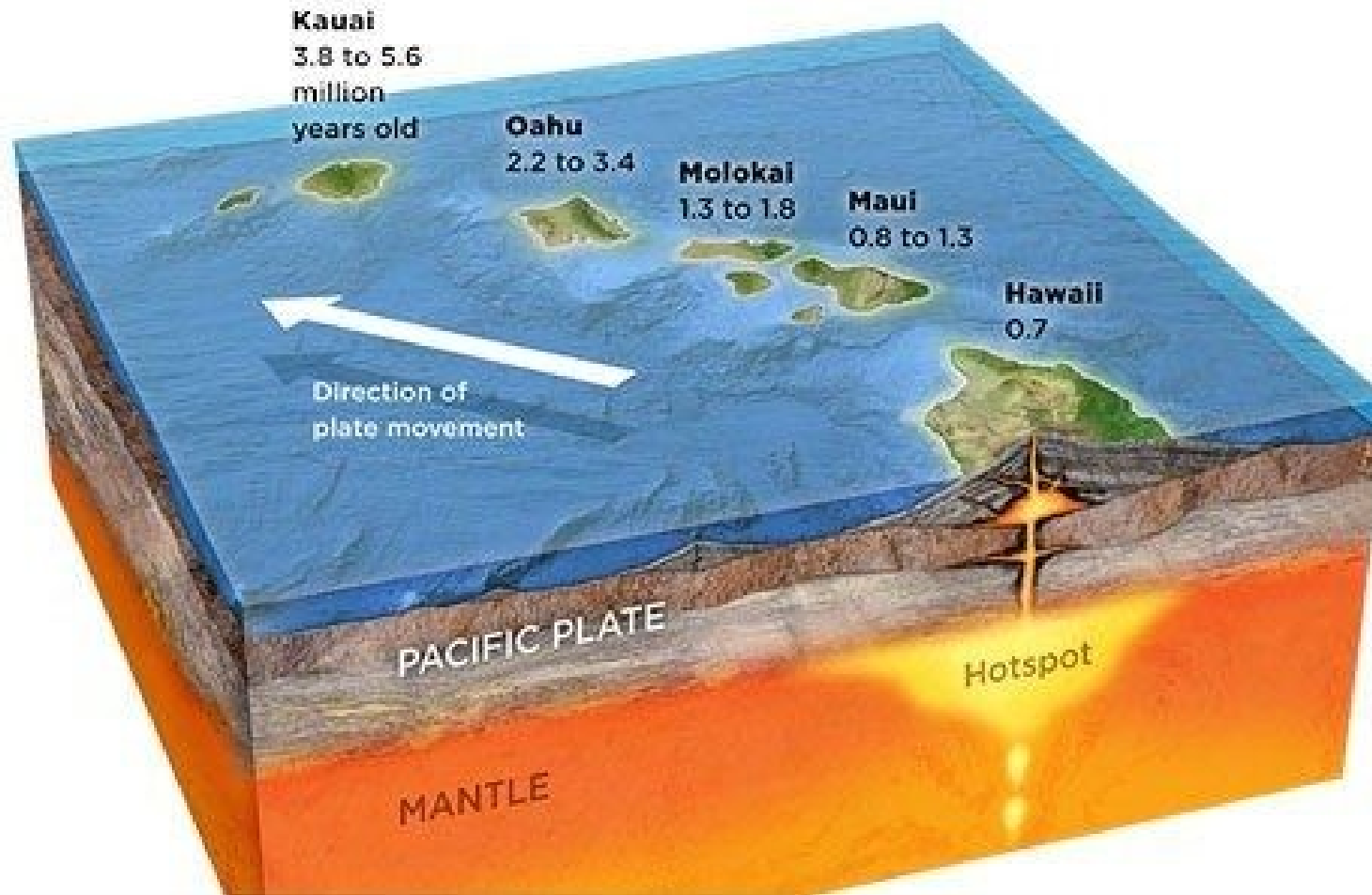
Aesthetic value: landscapes valued for their beauty

- This canyon is valued purely for its beauty - the colour of the rock, the curve of the river and the scale of the landscape draw visitors from around the world.
- Aesthetic value like this can also translate into economic value, through tourism.



Image: original resource, source listed in references.

Hawaii: a landscape built by a volcanic hotspot



The Hawaiian Islands formed one at a time as the Pacific Plate moved over a fixed volcanic hotspot.

Diagram: original resource, source listed in references.

Find out more: Hawaii's volcanic islands

Try these searches:

- "Hawaii volcanic hotspot formation video"
- "how the Hawaiian islands were formed"

Check these sites:

- abc.net.au/education
- geoscience.gov.au
- usgs.gov

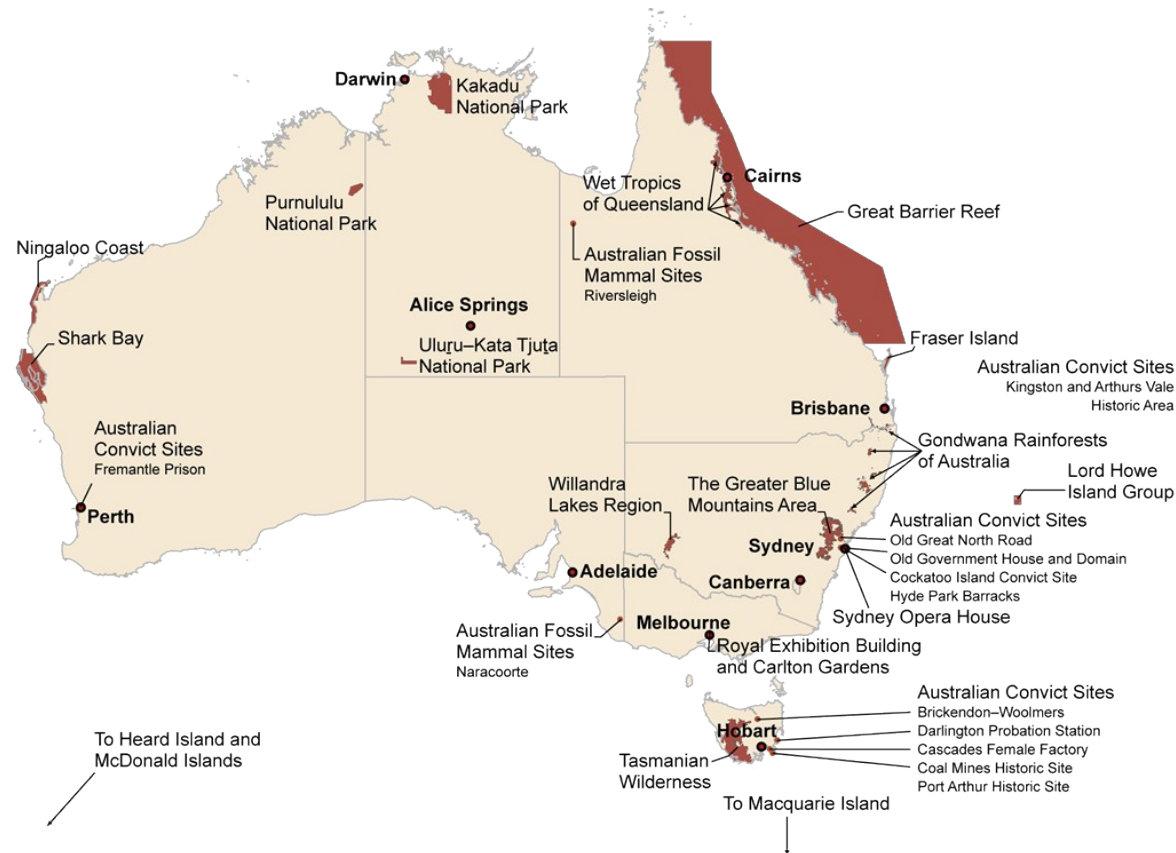
Over to you

List the locations in Australia that have aesthetic value to you.



Image: original resource, source listed in references.

World Heritage sites of Australia



Source: World Heritage Areas, Australia (2011), Environmental Resources Information Network, Australian Government
Department of Sustainability, Environment, Water, Population and Communities, based on Australian Coastline and
State Borders 1:100 000 (1990), Geoscience Australia

Australia's World Heritage listed sites and areas.

Source: World Heritage Areas, Australia (2011), Dept of Sustainability, Environment, Water, Population and Communities.

Over to you: World Heritage sites

World Heritage Site	Aesthetic or Cultural Value	Reason

Economic value of landscapes

How landscapes generate income for different people

Place and economic value of landscapes

Some landscapes hold economic significance. This can generate revenue through art and tourism.

1

TOURISM

Visitors travel specifically to see famous landscapes, spending money on travel, food and accommodation.

2

EMPLOYMENT

Tourism and land-based industries create local jobs.

3

INVESTMENT

Economic value can fund further conservation and infrastructure.

4

TRADE-OFFS

Economic use can clash with protecting a landscape's natural or cultural value.

Top tourist destinations in Australia



Image: original resource, source listed in references.

- Australia's landscapes are a major drawcard for both international and domestic tourists.
- Coastal destinations like this one depend on the beach, the water and the surrounding landscape staying attractive and accessible.

Case study: Uluru

- **Uluru is a large sandstone monolith that is 348 metres high and has a circumference of 9.4 km.**
- **Around 400,000 people visit Uluru every year, providing a large economic boost to the Anangu people, the land's traditional owners.**
- **Uluru carries deep cultural and spiritual value for the Anangu as well as economic value from tourism - both matter, and the two are managed together.**



Image: original resource, source listed in references.

Changing landscapes

How people use and change landscapes - the Australian coast

Changing landscapes in Australia

- Coastal areas are heavily changed in Australia. 80% of the population lives within 50 km of the coast.
- A clash between economic value, liveability and aesthetic value can develop as more people move to the coast.



Image: original resource, source listed in references.

Over to you

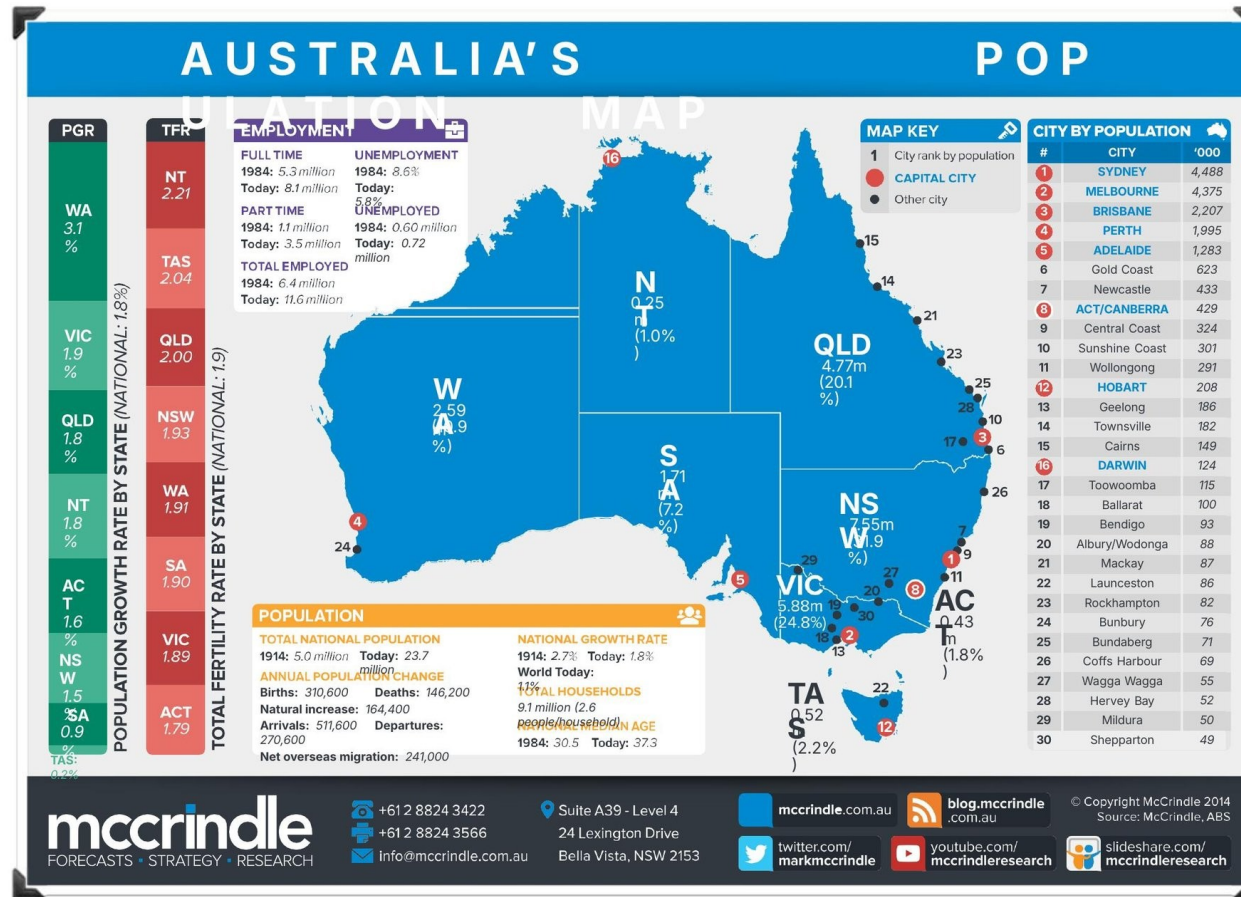
List the reasons why Australians live on the coast.



Image: original resource, source listed in references.

Think about climate, jobs, lifestyle and access to the water.

Australia's population: where people live

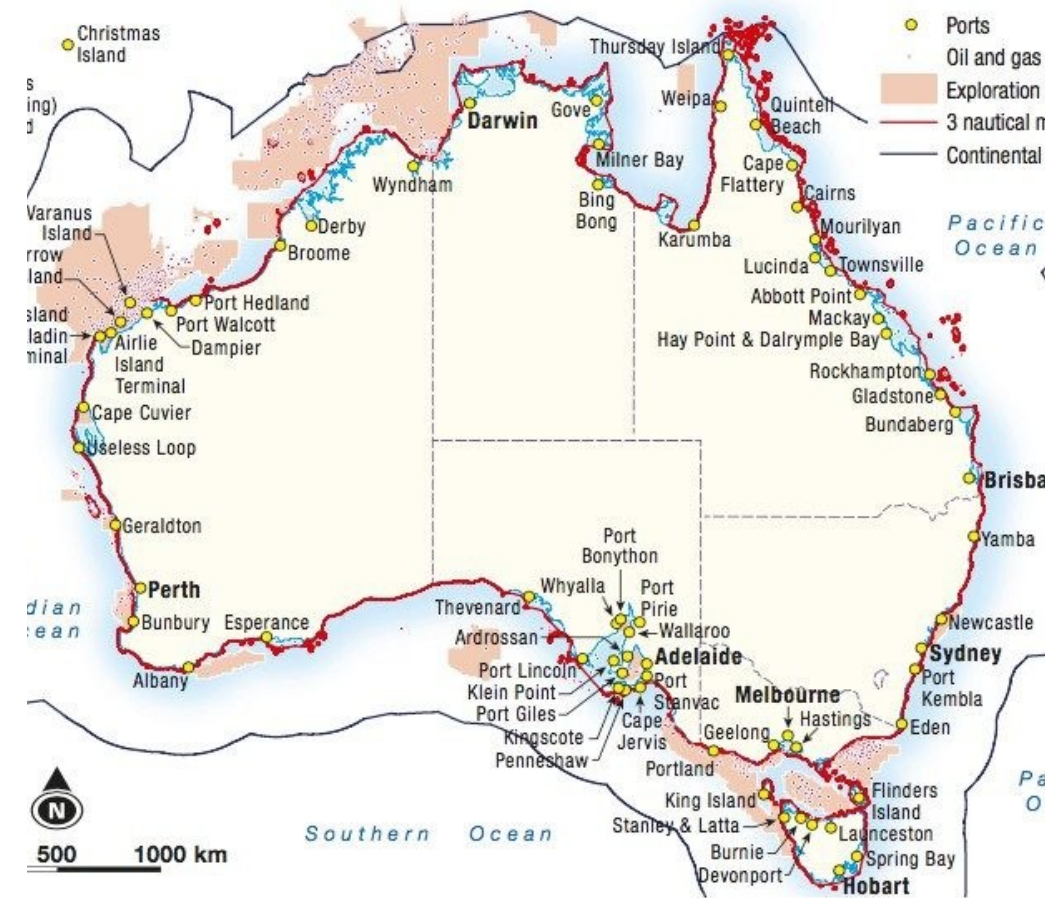


Most Australians live in the coastal south-east - NSW and Victoria together hold over half the population.

Source: McCrindle Research, "Australia's Population Map", based on ABS data.

Coastal issues for Australia

- Stormwater run-off and sewage flow into rivers, the sea and wetlands, threatening recreation, human health and ecosystems.
- Pesticides, fertilisers and other chemicals are released into stormwater or sewerage systems, increasing pollution.
- Coastal native bush and habitat is lost as city margins expand along the coast in a ribbon.
- Coastal wetlands are destroyed, leading to a loss of nursery grounds for fisheries.
- Toxic waste can be released from old coastal industrial sites or rubbish dumps during redevelopment.



3.10 Locations of oil, gas and ports in Australia

Text and map (Source 3.10): as featured in the original source deck - original textbook not independently verified.

References - sources used in this deck

Shipp, A. Landforms and Landscapes (Year 7 Geography, Topic 2) - original source deck for this resource

<https://www.slideshare.net/adeshipp/landforms-and-landscapes>

NASA. The Blue Marble (Apollo 17, 1972)

<https://visibleearth.nasa.gov/images/55418/blue-marble-the-earth-seen-from-apollo-17>

NASA Goddard Space Flight Center. Digital Tectonic Activity Map of the Earth

<https://denali.gsfc.nasa.gov/dtam/>

ABC Education. Landscapes and Landforms: Coastal landscapes

<https://www.abc.net.au/education/landscapes-and-landforms-coastal-landscapes/102317076>

Khan Academy. Introduction to plate tectonics

<https://www.khanacademy.org/science/middle-school-earth-and-space-science/x87d03b443efbea0a:the-geosphere/x87d03b443efbea0a:plate-tectonics/v/introduction-to-plate-tectonics>

Australian Government. World Heritage Areas, Australia (2011), Dept of Sustainability, Environment, Water, Population and Communities

<https://www.environment.gov.au/heritage/places/world-heritage-list>

McCrindle Research. Australia's Population Map, based on ABS data

<https://mccrindle.com.au>

Mapsofworld.com. New South Wales map

<https://www.mapsofworld.com>